

Experiment 'p_3_7_order' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_7_order

Natural language statement: Write a method that takes three integers and returns "in order" if they are sorted in ascending or descending order, or "not in order" otherwise.

Method signature: p_3_7_order (i: int, j: int, k: int) returns (s: string)

Ensures

- `if (i < j < k) || (i > j > k) then s == "in order" else s == "not in order"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes three integers and returns "in order" if they are sorted in
→ ascending or descending order, or "not in order" otherwise.

The signature should be:

`p_3_7_order (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if (i < j < k) || (i > j > k) then s == "in order" else s == "not in order"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdqaSmeTtOReue7IC8pghFKKbOd3

Model: gpt-5.5-2026-04-23

Created at: 1780485288

Finish reason: stop

Usage: 354 tokens in, and 208 tokens out

```
{
  if (i < j && j < k) || (i > j && j > k) {
    s := "in order";
  } else {
    s := "not in order";
  }
}
```

Final Program

```
method p_3_7_order (i: int, j: int, k: int) returns (s: string)
  ensures if (i < j < k) || (i > j > k) then s == "in order" else s == "not in order"
  → "
{
  if (i < j && j < k) || (i > j && j > k) {
    s := "in order";
  } else {
    s := "not in order";
  }
}
```

```
}  
}
```

Total Token Usage

Input tokens: 354

Output tokens: 208

Reasoning tokens: 138

Sum of ‘total tokens’: 562

Experiment Timings

Overall Experiment started at 1780485288103, ended at 1780485296583, lasting 8480ms (8.48 seconds)

Iteration #1 started at 1780485288104, ended at 1780485296583, lasting 8479ms (8.48 seconds)